

# Nectar Data Scientist Challenge

## Position

**Role:** Data Scientist

**Experience:** 1–3 Years

**Location:** Coimbatore, Tamil Nadu, India

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## About Nectar

Nectar is an AI-first product company building intelligent platforms that transform how enterprises leverage AI, automation, and connected technologies.

Our platform combines Artificial Intelligence, IoT, Agentic AI, and cloud-native technologies to deliver intelligent, scalable, and secure experiences for customers worldwide.

As a Data Scientist at Nectar, you will work on real-world challenges involving connected assets, sensor telemetry, predictive analytics, and operational intelligence.

This assessment is designed to evaluate your analytical thinking, machine learning expertise, and ability to convert IoT data into actionable product capabilities.

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## Challenge Overview

You are working on Nectar's Intelligent Facilities Platform.

A customer operates multiple commercial buildings, each containing several interconnected assets such as:

- HVAC Systems
- Chillers
- Pumps
- Air Handling Units (AHUs)
- Energy Meters
- Environmental Sensors

Thousands of sensors continuously stream telemetry data from these assets.

Your objective is to analyze historical IoT data, understand asset relationships, identify operational inefficiencies, and build intelligent capabilities that can improve reliability, reduce downtime, and optimize energy consumption.

# Datasets Provided

You will receive the following datasets.

## 1. Sensor Telemetry Dataset

Historical sensor readings collected from connected assets.

| Column            | Description                      |
|-------------------|----------------------------------|
| timestamp         | Timestamp of sensor reading      |
| site_id           | Unique site identifier           |
| building_id       | Building identifier              |
| asset_id          | Unique asset identifier          |
| temperature       | Temperature reading (°C)         |
| humidity          | Humidity (%)                     |
| pressure          | Pressure reading                 |
| vibration         | Equipment vibration level        |
| power_consumption | Energy consumption (kWh)         |
| occupancy_count   | Number of occupants              |
| operating_mode    | Cooling/Heating/Idle             |
| fault_flag        | Historical fault indicator (0/1) |

## 2. Asset Metadata Dataset

Information describing deployed assets.

| Column            | Description               |
|-------------------|---------------------------|
| asset_id          | Unique asset identifier   |
| site_id           | Site identifier           |
| asset_name        | Asset name                |
| asset_type        | Chiller/AHU/Pump/etc.     |
| manufacturer      | Manufacturer              |
| installation_date | Installation date         |
| capacity          | Rated capacity            |
| parent_asset_id   | Parent asset relationship |

## 3. Asset Connectivity Dataset

Represents relationships between assets.

| Column                | Description                |
|-----------------------|----------------------------|
| source_asset_id       | Parent asset               |
| target_asset_id       | Connected asset            |
| connection_type       | Supplies/Controls/Monitors |
| relationship_strength | Connectivity weight        |

# Tasks

## Task 1: Exploratory Data Analysis (EDA)

Perform a detailed exploratory analysis on the telemetry data.

Your analysis should:

- Understand data distributions.
- Identify missing values and anomalies.
- Discover temporal patterns.
- Analyze asset behavior across sites.
- Compare performance across asset types.
- Identify factors influencing equipment failures.
- Identify factors affecting energy consumption.

## Deliverables

Provide:

- Visualizations
  - Statistical summaries
  - Key observations
  - Business insights
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## Task 2: Predictive Maintenance

Develop a machine learning model to predict whether an asset is likely to fail within the next 24 hours.

The model should help operations teams proactively schedule maintenance activities.

## Expected Activities

- Data preprocessing
- Feature engineering
- Handling missing values
- Model selection

- Model training
- Model evaluation
- Error analysis

## Suggested Algorithms

- Random Forest
- XGBoost
- LightGBM
- CatBoost
- Any appropriate approach

## Evaluation Metrics

- Precision
- Recall
- F1 Score
- ROC-AUC

## Explain

- Why the selected model was chosen.
  - Important features driving predictions.
  - Potential business impact.
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# Task 3: Energy Consumption Forecasting

Develop a forecasting model to predict energy consumption for the next 24 hours for each building.

The goal is to support proactive energy optimization.

## Suggested Techniques

- Prophet
- ARIMA
- XGBoost
- LSTM
- Temporal Fusion Transformer

- Any suitable approach

## Evaluation Metrics

- MAE
- RMSE
- MAPE

## Expected Output

- Forecast plots
  - Model evaluation
  - Key insights
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# Task 4: Anomaly Detection

Design an anomaly detection framework capable of identifying abnormal equipment behavior.

Examples include:

- Sudden power spikes
- Excessive vibration
- Sensor drift
- Abnormal temperature patterns
- Equipment degradation

## Suggested Techniques

- Isolation Forest
- Autoencoders
- Statistical Thresholding
- Clustering
- Change Point Detection

## Deliverables

- Anomaly detection methodology
- Visualizations
- Identified anomalies

- Business recommendations
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## Task 5: Multi-Asset Connectivity Analysis

Each customer site contains multiple interconnected assets.

For example:

Site A

- Chiller-01
  - AHU-01
  - AHU-02
- Pump-01
  - Flow Sensor-01
  - Energy Meter-01

Your objective is to model and analyze relationships between assets.

## Requirements

### Build Asset Hierarchy

Create a hierarchical representation of assets within each site.

Examples:

- Site → Building → Floor → Asset
  - Chiller → AHU → Sensor
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### Connectivity Analysis

Analyze relationships among assets and answer:

- Which assets are connected?
  - Which assets depend on each other?
  - Which assets are isolated or disconnected?
  - What downstream assets are affected if a parent asset fails?
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## Failure Impact Analysis

Assume a critical asset fails.

Examples:

- Chiller failure
- Pump failure

Determine:

- Impacted downstream assets
  - Potential operational impact
  - Recommended mitigation strategies
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## Data Quality Assessment

Identify:

- Missing relationships
  - Duplicate connections
  - Orphan assets
  - Invalid parent-child mappings
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## Expected Deliverables

Provide:

- Asset hierarchy visualization
  - Dependency graph/network visualization
  - Connectivity analysis report
  - Failure propagation analysis
  - Data quality findings
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## Bonus

Use graph-based techniques such as:

- GraphQL

Implement example queries such as:

- Show all assets connected to Chiller-01.
  - Identify all downstream assets impacted by AHU-03 failure.
  - List all assets under Site A.
  - Find isolated assets with no parent or child relationships.
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## Bonus Challenge (Optional)

Complete any one of the following.

### Option A: Interactive Dashboard

Develop an interactive dashboard using:

- Streamlit
- Power BI
- Tableau

Dashboard should include:

- Site overview
  - Asset health status
  - Failure predictions
  - Energy trends
  - Anomaly alerts
  - Asset connectivity visualization
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## Model Deployment

Deploy one machine learning model as an API using:

- FastAPI
- Flask

The API should expose prediction endpoints.

Example:

POST /predict\_failure

Input:

Asset telemetry data

Output:

Failure probability

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## Deliverables

Please submit:

1. Source Code (GitHub Repository preferred)
  2. Jupyter Notebook(s)
  3. README containing:
    - Setup instructions
    - Architecture overview
    - Assumptions
    - Design decisions
  4. Report or Presentation (Maximum 5 pages/slides)
  5. Optional deployment/demo link
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## Evaluation Criteria

| Criteria              | Weightage |
|-----------------------|-----------|
| Problem Understanding | 10%       |
| Exploratory Analysis  | 15%       |
| Feature Engineering   | 10%       |
| Predictive Modeling   | 20%       |
| Forecasting           | 10%       |
| Anomaly Detection     | 10%       |

Asset Connectivity Analysis 15%

Code Quality & Documentation 5%

Communication & Presentation 5%

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## Submission Timeline

Please submit your completed assignment within **5 calendar days** from the date of receiving the challenge.

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## Important Notes

- Clearly document assumptions wherever applicable.
- Focus on explainability and business impact.
- Production-quality code and reproducibility are highly valued.
- Creativity and experimentation are encouraged.

We are interested not only in the final solution but also in understanding your thought process, trade-offs, and engineering decisions.

We look forward to reviewing your submission.

**Team Nectar**